User Provisioning

Provisioning is the process of creating, maintaining and deleting digital identities (accounts) for users of a system(s) and linking appropriate rights to identities in the form of rules and roles. Identity provisioning is key for the Identity Federation. Identity federation is a mechanism that allows authentication across different enterprises in different trust domains based on a trust factor. This makes access easy, as users do not have to remember a different set of credentials for every application they use. See the following identity provisioning key concepts that are used in WSO2 Identity Server.

- Inbound provisioning

Inbound provisioning provisions users or groups into the WSO2 Identity Server by an external application. These external applications are referred to as service providers. WSO2 Identity Server supports the SCIM 2.0 API standards for inbound provisioning.

- Outbound provisioning

Outbound provisioning provisions users to a trusted identity provider from the WSO2 Identity Server. A trusted identity provider is basically an identity provider that supports inbound provisioning. It can be Google, Salesforce, another Identity Server, etc.

This tutorial guides you to perform inbound provisioning using SCIM2, and outbound provisioning and Just-In-Time provisioning using two WSO2 Identity Servers.

Inbound provisioning Using SCIM2

Setting up:

- Download and start the WSO2 Identity Server 7.3.0

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In the requests below, the Authorization header uses **YWRtaW46YWRtaW4=**, which is admin:admin encoded in Base64. To encode other credentials, run: echo -n '<username>:<password>' | base64

Try it :

1. Provision the user into the Identity Server

Request

```
curl -X 'POST' \
      'https://localhost:9443/scim2/Users' \
      -H 'accept: application/scim+json' \
      -H 'Content-Type: application/scim+json' \
      -H 'Authorization: Basic YWRtaW46YWRtaW4=' \

      -d '{
      "schemas": [],
      "name": {
        "givenName": "Kim",
        "familyName": "Berry"
      },
      "emails": [
        {
            "value": "kim@wso2.com",
            "primary": true
        }
      ],
      "userName": "kim",
      "password": "MyPa33w@rd"
}'
```

Response

```
{"emails":["kim@wso2.com"],"meta":
{"created":"2024-04-24T14:03:28.890629Z","location":"https://
localhost:9443/scim2/Users/8ade213f-f3ad-4e49-
b975-18b379ded76b","lastModified":"2024-04-24T14:03:28.890629Z","res
ourceType":"User"},"schemas":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:scim:
schemas:extension:enterprise:2.0:User","urn:scim:wso2:schema"],"role
s":[{"audienceValue":"10084a8d-113f-4211-a0d5-
efe36b082211","display":"everyone","audienceType":"organization","va
lue":"8f38e212-cd6a-4027-9244-f851a81589e4","$ref":"https://
localhost:9443/scim2/v2/Roles/8f38e212-cd6a-4027-9244-
```

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```
f851a81589e4","audienceDisplay":"Super"}],"name":
{"givenName":"Kim","familyName":"Berry"},"id":"8ade213f-f3ad-4e49-
b975-18b379ded76b","userName":"kim"}
```


2. View the provisioned users using user id

Obtain the user id from the above response.

Request

```
curl -X 'GET' \
'https://localhost:9443/scim2/Users/{user-id}' \
-H 'accept: application/scim+json' \
-H 'Authorization: Basic YWRtaW46YWRtaW4='
```

3. Update the provisioned user details

In response you will get the updated user profile.

Request

```
curl -X 'PUT' \
'https://localhost:9443/scim2/Users/{user-id}' \
-H 'accept: application/scim+json' \
-H 'Content-Type: application/scim+json' \
-H 'Authorization: Basic YWRtaW46YWRtaW4='
-d '{
"schemas": [],
"name": {
  "givenName": "Jimmy",
  "familyName": "Kimmel"
},
"userName": "kim",
"emails": [
  {
      "value": "kim@test.com",
      "primary": true
  }
],
}'
```

Response

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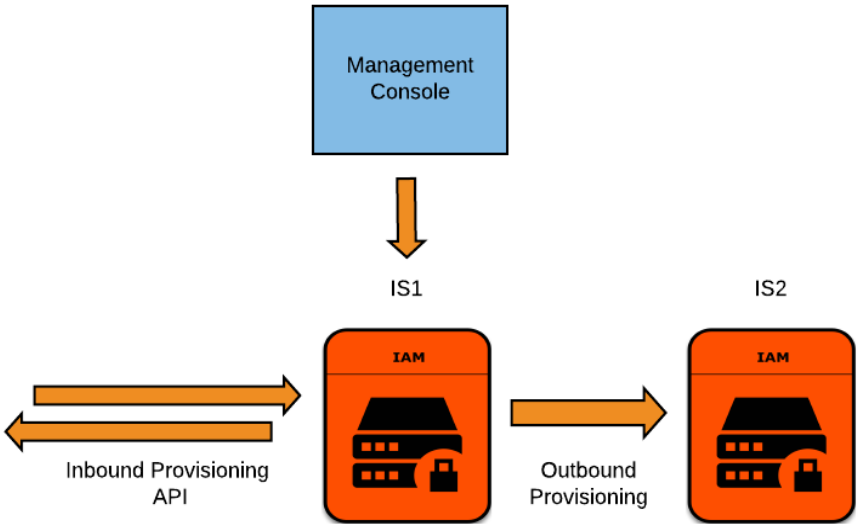
```
{"emails":["kim@test.com"],"meta":
{"created":"2024-04-24T14:03:28.890629Z","location":"https://
localhost:9443/scim2/Users/8ade213f-f3ad-4e49-
b975-18b379ded76b","lastModified":"2024-04-24T14:04:41.567169Z","
resourceType":"User"},"schemas":
["urn:ietf:params:scim:schemas:core:2.0:User","urn:ietf:params:sc
im:schemas:extension:enterprise:2.0:User","urn:scim:wso2:schema"]
,"roles":[{"audienceValue":"10084a8d-113f-4211-a0d5-
efe36b082211","display":"everyone","audienceType":"organization",
"value":"8f38e212-cd6a-4027-9244-f851a81589e4","$ref":"https://
localhost:9443/scim2/v2/Roles/8f38e212-cd6a-4027-9244-
f851a81589e4","audienceDisplay":"Super"}],"name":
{"givenName":"Kim","familyName":"Berry"},"id":"8ade213f-
f3ad-4e49-b975-18b379ded76b","userName":"kim"}
```


4. Delete the provisioned user by user id

Request

```
curl -X 'DELETE' \
'https://localhost:9443/scim2/Users/{user-id}' \
-H 'accept: */*' \
-H 'Authorization: Basic YWRtaW46YWRtaW4='
```

Outbound Provisioning



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In this section, we configure one Identity Server to provision users to another Identity Server using SCIM.

For ease, let's name the WSO2 Identity Server doing JIT provisioning as **IS1** and the identity provider WSO2 Identity Server as **IS2**.

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Prerequisite: Configuring port offset on IS2

To run two IS instances at the same time, the port of one must be changed from the default value of 9443. To change the port of IS2 to 9444, do the following,

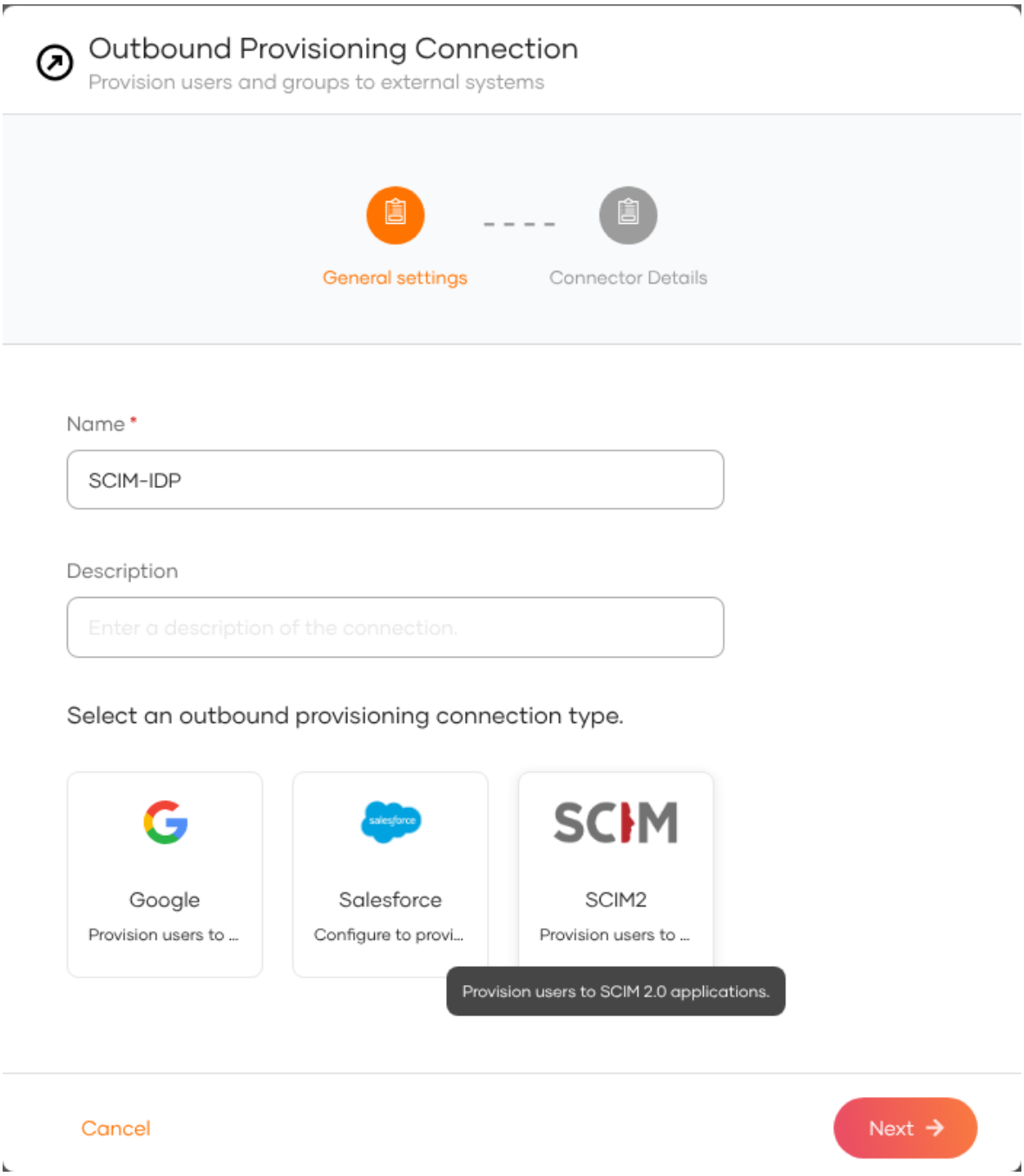
1. Navigate to `<IS2_HOME>/repository/conf` folder. Open the `deployment.toml` file in a text editor.
2. Add `offset = 1` under the `[server]` section and save the file.
3. Start IS2.

Setting up:

Step 1: Configuring an identity provider

1. Login to the WSO2 Identity Server **Console** of **IS1,** using your admin credentials (e.g. admin:admin).
2. In the WSO2 Identity Server **Console,** from the menu click **Connections.**
3. Click **+ New Connection**, and on the **Outbound Provisioning Connection** card click **Create**.

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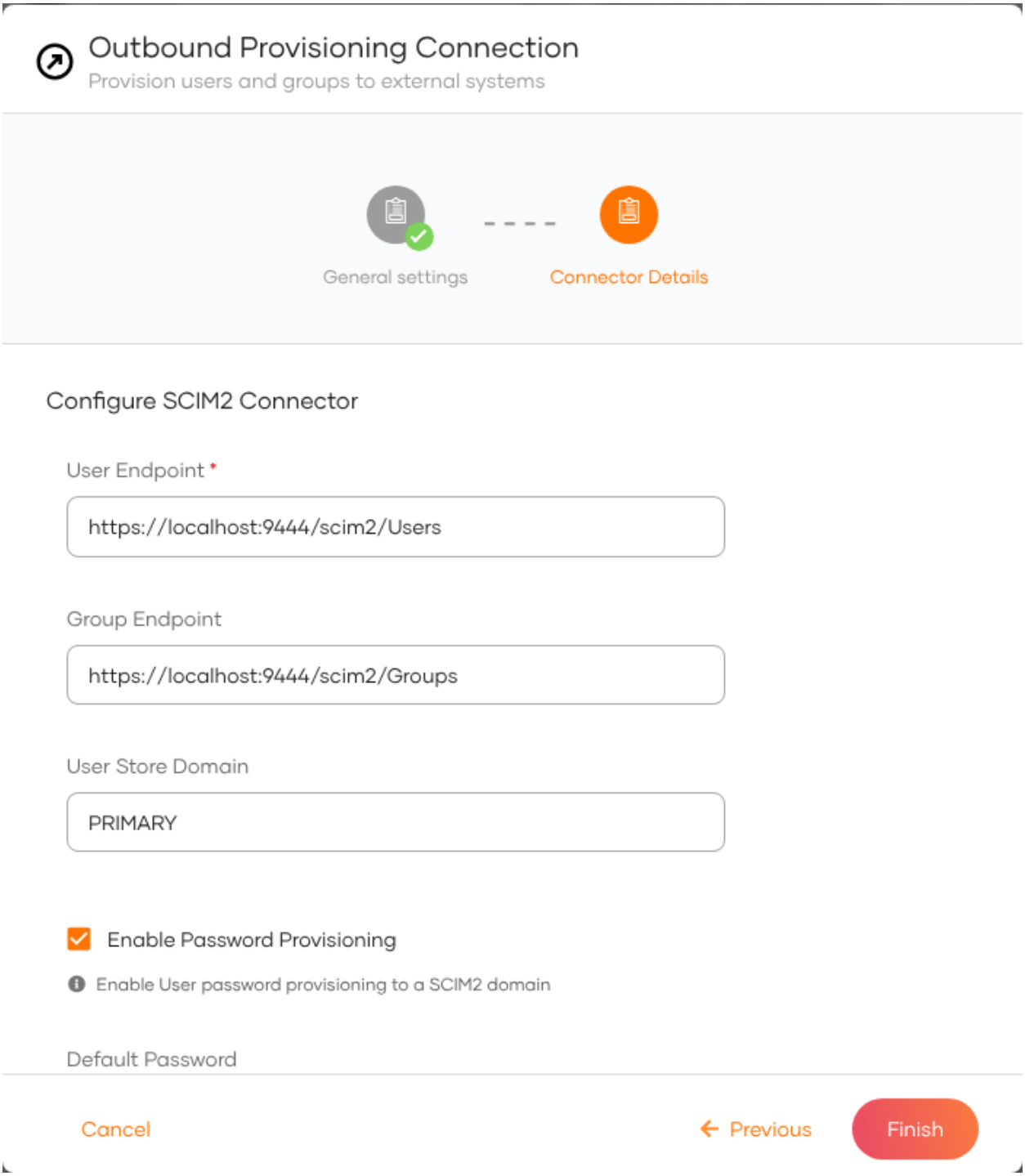
4. Enter a unique name for the connection (for example, **SCIM-IDP**), select **SCIM2** as the outbound provisioning connection type and click **Next**.
5. Enter the following values in the respective fields (keep **Authentication Mode** as **Basic Authentication**) and click **Finish**.

Field	Value

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Username	Username of SCIM application (in this case, a username of a user registered in IS2, such as “admin”)
Password	Password of the user entered in the username field
User Endpoint	https://localhost:9444/scim2/Users
Group Endpoint	https://localhost:9444/scim2/Groups
User Store Domain	PRIMARY
Enable Password Provisioning	Keep this checked.

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6. The connection is created with the SCIM2 connector already enabled, so no further action is needed on this page.

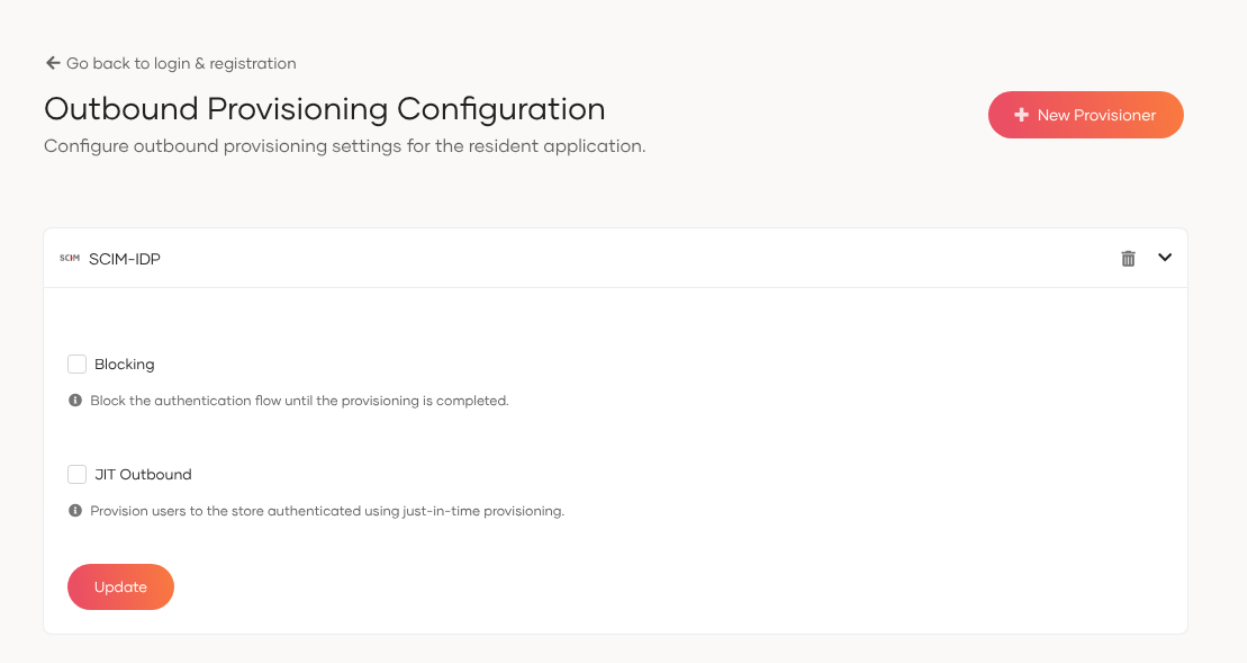
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Step 2: Configuring the resident service provider

When configuring outbound provisioning for any user management operation done via the console, outbound provisioning identity providers must be configured against the resident service provider. So, based on the outbound configuration, users added from the console can also be provisioned to external systems, in this case, IS2.

1. Login to the WSO2 Identity Server **Console** of **IS1,** using your admin credentials (e.g. admin:admin).
2. In the WSO2 Identity Server **Console,** from the menu click **Login & Registration.**
3. Go to **Provisioning Settings** > **Outbound Provisioning Configuration** and click **New Provisioner**.
4. Select the connection in which you have configured outbound provisioning (**SCIM-IDP**) as the **Provisioning Connection**. Leave **Blocking** and **JIT Outbound** unticked.
5. Click **Finish**.

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Go back to login & registration
Outbound Provisioning Configuration
Configure outbound provisioning settings for the resident application.
New Provisioner
SCIM-IDP
Blocking
Block the authentication flow until the provisioning is completed.
JIT Outbound
Provision users to the store authenticated using just-in-time provisioning.
Update

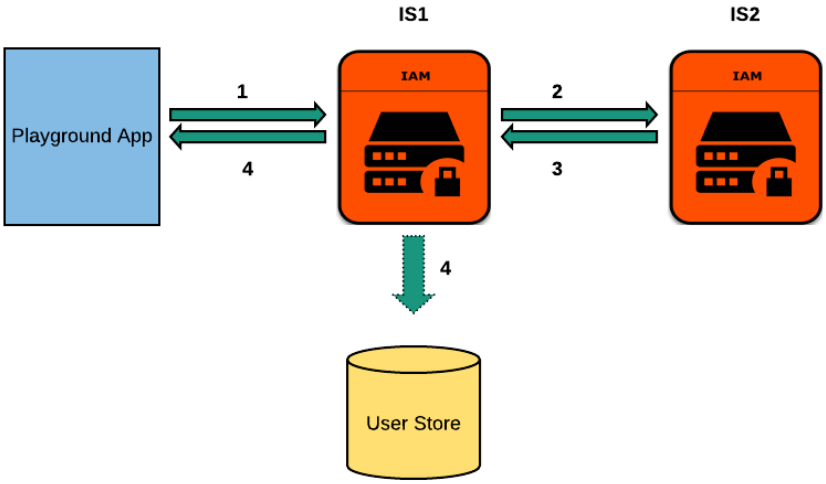
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Try it

1. Log in to the **Console** of **IS1**.
2. Add a new user.
    a. Expand the **User Management** menu item, click on **Users**.
    b. On the **Users page**, click **Add User** and select **Single User** from the dropdown.
    c. Fill all required fields and click **Finish**.
3. Log in to the management console of IS2.
4. Expand the **User Management** menu item, click on **Users**.
5. You will observe that the user added to IS1 has been added to the IS2 userstore as well.

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Just-In-Time Provisioning



JIT provisioning is a means of provisioning users into the Identity Server at the time of federated authentication. A service provider initiates the authentication request, the user gets redirected to the Identity Server, and then the Identity Server redirects the user to an external identity provider for authentication. JIT provisioning gets triggered only when the Identity Server receives a positive authentication response from the external identity provider. The Identity Server will provision the user to its internal user store with the user claims from the authentication response.

In this section, we discuss how to set up one Identity Server for JIT provisioning from another Identity Server acting as an identity provider.

For ease, let's name the WSO2 Identity Server doing JIT provisioning as **IS1** and the identity provider WSO2 Identity Server as **IS2**.

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Setting Up

Step 1: Configure a service provider on IS2

1. Start IS2.
2. Login to the WSO2 Identity Server **Console and** from the menu click **Applications.**
3. Click **New Application.**
4. From the given set of templates select **Traditional Web Application** and select **OpenID Connect** as the protocol.
5. Add name as SP-IS2 and Authorized redirect URLs as https://localhost:9443/commonauth.
6. Click **Create**.
7. Make a note of the **Client ID** and **Client Secret** of this Service Provider.

Step 2: Configuring IS2 as an identity provider with JIT provisioning in IS1

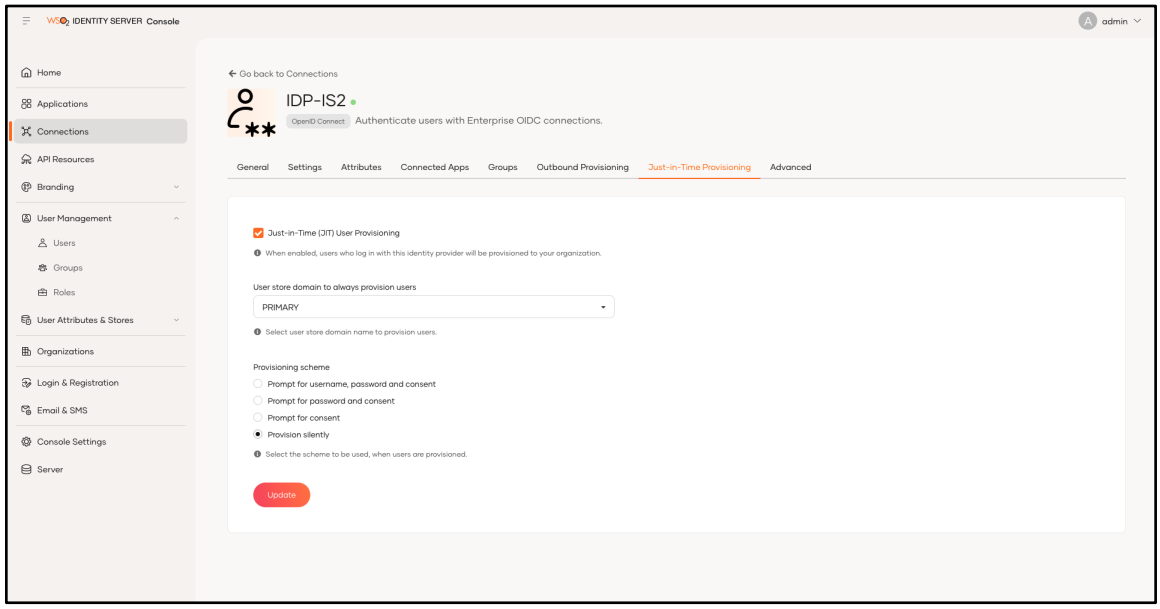
1. Start IS1.
2. Login to the WSO2 Identity Server **Console,** from the menu click **Connections.**
3. Click + **New Connection**.
4. From the available templates select **Standard-Based IdP** and click **Create**.
5. Enter **IDP-IS2** as the Identity provider name, select **OpenID Connect** as the protocol and click **Next**.
6. Fill the following fields and click **Next**:

Field	Value
Client ID	Enter the client ID of the service provider registered in IS2

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Client Secret	Enter the client secret of the service provider registered in IS2
Authorization Endpoint URL	https://localhost:9444/oauth2/authorize
Token Endpoint URL	https://localhost:9444/oauth2/token

7. On the **Certificates** step, click **Finish**. Then open the **Just-in-Time Provisioning** tab of the new connection.
8. **Check** Just-in-Time (JIT) User Provisioning to enable JIT provisioning.



9. Select **PRIMARY** in the dropdown to provision users to the primary userstore, and then select **Provision silently** to allow the provisioning process to complete without prompting for extra details (for demonstration purposes).
10. Click **Update**.

Step 3: Configuring a service provider in IS1 to federate authentication to IS2

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We will be using the playground2 sample app to test the flow. Copy `playground2.war` (in the `3. Resources` folder of Day 12) into `<TOMCAT_HOME>/webapps` and make sure Tomcat is running: from `<TOMCAT_HOME>/bin` run `catalina.bat start` on Windows or `sh catalina.sh start` on Linux and macOS (see Lab 05 for the full Tomcat setup).

1. In the WSO2 Identity Server **Console,** from the menu click **Applications.**
2. Click **New Application.**
3. From the given set of templates select **Traditional Web Application** template.
4. Fill the fields in the create application wizard.

   Name: *playground2*

   Protocol: *OpenID Connect*

   Authorized redirect URLs: http://localhost:8080/playground2/oauth2client
5. Make a note of the **Client ID** of this registered OAuth/OpenID Connect service provider.
6. Click on the **Login Flow tab**.
7. Add a new sign in option to the first step. Click on **+Add Sign In Option** Button.
8. Select **IDP-IS2** and click **Update**.

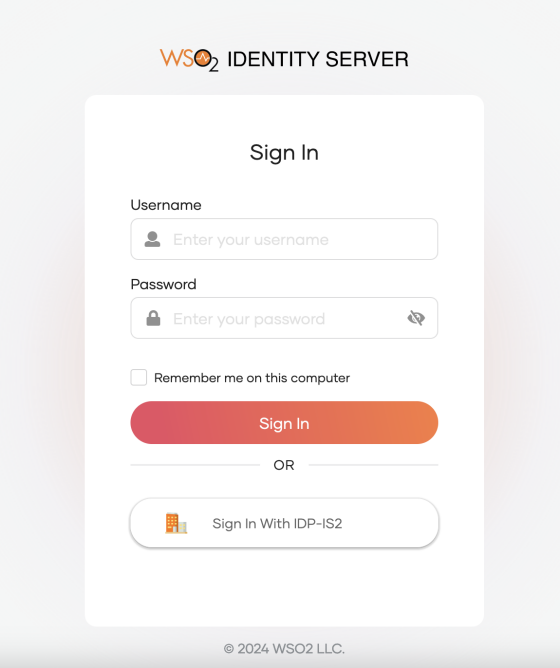
Step 4: Add a user in IS2

1. Log in to the management console of IS2.
2. Add a new user.
   a. Expand the **User Management** menu item, click on **Users**.
   b. On the **Users page**, click **Add User** and select **Single User** from the dropdown.
   c. Fill all required fields and click **Finish**.

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Try it

1. Access the playground2 app via http://localhost:8080/playground2/oauth2.jsp
2. Enter the client ID of the service provider registered in IS1 in the **Client ID** field.
3. Click **Authorize**. The browser will be redirected to the login page of IS1.
4. From that login page select the IDP-IS2 option to authenticate users from the IS2.



5. Enter the credentials of the user registered in IS2 and click **Login**.
6. Once the authentication is successful, log in to the management console of IS1 and list the users. You will observe that the user registered in IS2 has now been added to the userstore of IS1 as well.

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